

Identifying Known Acts in Document Chunks

Exploring matching strategies for C1 positive chunk identification

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Motivation

The C1 codebook classifies whether a document chunk contains a substantive description of a fiscal measure. To evaluate C1 via LOOCV, we need to know which chunks are **positive** (contain a known fiscal act from [aligned_data](#)) and which are **negative** (contain no fiscal policy discussion).

Our current tier system identifies positives at two levels:

- **Tier 1:** Chunk contains verbatim passage text from Romer and Romer (2010) labels
- **Tier 2:** Chunk contains exact act name match with fiscal keyword co-occurrence

The question is whether these criteria are too strict, causing false negatives (chunks that genuinely discuss a known act but are missed by our matching). This notebook investigates the problem and explores solutions.

Current State: Tier Distribution

Table 1: Current Chunk Tier Distribution

Tier	Chunks	Percent
Tier 1	70	0.5%
Tier 2	1986	15.1%
Negative	11632	88.7%
Total	13112	100.0%

The problem: 80% of chunks are excluded (gray zone) because they match `relevance_keys` but don't match any act name. Only 35 chunks qualify as clean negatives (1.2%). This means precision is essentially untestable, and many chunks that discuss known acts may be missed.

Why Passages Don't Help with Alias Discovery

One natural idea is to extract alternative act names from the Romer and Romer (2010) passage text. But these passages are presidential speeches and committee testimony selected to illustrate each act's **motivation**, not to name the legislation. They rarely name the act itself.

Table 2: Do R&R Passages Name the Act?

Check	Result
Acts where passages contain exact full name	8 / 39
Acts where passages contain short name (no year)	8 / 39
Acts where passages contain neither	31 / 39

The passages use generic fiscal vocabulary ("tax cut", "tax relief", "deficit reduction"), not legislative names. This rules out extracting aliases from passage text.

OCR Line Breaks: The Primary Matching Problem

Government PDFs from the 1940s-1990s introduce line breaks within act names during OCR extraction. For example, "Revenue Act of 1962" may appear as:

Revenue Act of

Revenue Act

Revenue of 1962

Current Tier 2 uses `str_detect(text, fixed(act_name))` which requires the name to appear as a contiguous string. This fails on OCR-broken names.

Test: Does whitespace normalization help?

We compare current Tier 2 matching (exact `fixed()` on raw text) against matching after `str_squish()` collapses all whitespace.

Table 3: Impact of Whitespace Normalization on Tier 2 Matching

Metric	Value
Total Tier 2 matches (current)	1,219
Total Tier 2 matches (squished)	1,738
Additional matches recovered	+519 (43% increase)
Acts with 0 matches (current)	5
Acts with 0 matches (squished)	5

Table 4: Per-Act Impact of Whitespace Normalization

Act	Year	Current	Squished	Gained
Economic Growth and Tax Relief Reconciliation Act of 2001	2001	24	111	87
Jobs and Growth Tax Relief Reconciliation Act of 2003	2003	24	106	82
Omnibus Budget Reconciliation Act of 1990	1990	30	67	37
Tax Reform Act of 1986	1986	79	111	32
Tax Equity and Fiscal Responsibility Act of 1982	1982	28	56	28
Job Creation and Worker Assistance Act of 2002	2002	14	40	26
Omnibus Budget Reconciliation Act of 1993	1993	45	70	25
Economic Recovery Tax Act of 1981	1981	86	107	21
Revenue Act of 1950	1950	55	71	16
Social Security Amendments of 1983	1983	52	66	14
Tax Reform Act of 1969	1969	64	77	13
Crude Oil Windfall Profit Tax Act of 1980	1980	25	37	12
Reconciliation Budget Act of 1989	1989	73	84	11
Revenue Act of 1950	1950	55	71	16
Revenue Act of 1968	1968	55	71	16
Revenue Act of 1978	1978	73	84	11
Revenue Act of 1981	1981	86	107	21
Revenue Act of 1986	1986	79	111	32
Revenue Act of 1990	1990	30	67	37
Revenue Act of 1993	1993	45	70	25
Revenue Act of 2001	2001	24	111	87
Revenue Act of 2003	2003	24	106	82
Revenue Act of 2002	2002	14	40	26
Revenue Act of 2004	2004	14	40	26
Revenue Act of 2005	2005	14	40	26
Revenue Act of 2006	2006	14	40	26
Revenue Act of 2007	2007	14	40	26
Revenue Act of 2008	2008	14	40	26
Revenue Act of 2009	2009	14	40	26
Revenue Act of 2010	2010	14	40	26
Revenue Act of 2011	2011	14	40	26
Revenue Act of 2012	2012	14	40	26
Revenue Act of 2013	2013	14	40	26
Revenue Act of 2014	2014	14	40	26
Revenue Act of 2015	2015	14	40	26
Revenue Act of 2016	2016	14	40	26
Revenue Act of 2017	2017	14	40	26
Revenue Act of 2018	2018	14	40	26
Revenue Act of 2019	2019	14	40	26
Revenue Act of 2020	2020	14	40	26
Revenue Act of 2021	2021	14	40	26
Revenue Act of 2022	2022	14	40	26
Revenue Act of 2023	2023	14	40	26
Revenue Act of 2024	2024	14	40	26
Revenue Act of 2025	2025	14	40	26
Revenue Act of 2026	2026	14	40	26
Revenue Act of 2027	2027	14	40	26
Revenue Act of 2028	2028	14	40	26
Revenue Act of 2029	2029	14	40	26
Revenue Act of 2030	2030	14	40	26
Revenue Act of 2031	2031	14	40	26
Revenue Act of 2032	2032	14	40	26
Revenue Act of 2033	2033	14	40	26
Revenue Act of 2034	2034	14	40	26
Revenue Act of 2035	2035	14	40	26
Revenue Act of 2036	2036	14	40	26
Revenue Act of 2037	2037	14	40	26
Revenue Act of 2038	2038	14	40	26
Revenue Act of 2039	2039	14	40	26
Revenue Act of 2040	2040	14	40	26
Revenue Act of 2041	2041	14	40	26
Revenue Act of 2042	2042	14	40	26
Revenue Act of 2043	2043	14	40	26
Revenue Act of 2044	2044	14	40	26
Revenue Act of 2045	2045	14	40	26
Revenue Act of 2046	2046	14	40	26
Revenue Act of 2047	2047	14	40	26
Revenue Act of 2048	2048	14	40	26
Revenue Act of 2049	2049	14	40	26
Revenue Act of 2050	2050	14	40	26
Revenue Act of 2051	2051	14	40	26
Revenue Act of 2052	2052	14	40	26
Revenue Act of 2053	2053	14	40	26
Revenue Act of 2054	2054	14	40	26
Revenue Act of 2055	2055	14	40	26
Revenue Act of 2056	2056	14	40	26
Revenue Act of 2057	2057	14	40	26
Revenue Act of 2058	2058	14	40	26
Revenue Act of 2059	2059	14	40	26
Revenue Act of 2060	2060	14	40	26
Revenue Act of 2061	2061	14	40	26
Revenue Act of 2062	2062	14	40	26
Revenue Act of 2063	2063	14	40	26
Revenue Act of 2064	2064	14	40	26
Revenue Act of 2065	2065	14	40	26
Revenue Act of 2066	2066	14	40	26
Revenue Act of 2067	2067	14	40	26
Revenue Act of 2068	2068	14	40	26
Revenue Act of 2069	2069	14	40	26
Revenue Act of 2070	2070	14	40	26
Revenue Act of 2071	2071	14	40	26
Revenue Act of 2072	2072	14	40	26
Revenue Act of 2073	2073	14	40	26
Revenue Act of 2074	2074	14	40	26
Revenue Act of 2075	2075	14	40	26
Revenue Act of 2076	2076	14	40	26
Revenue Act of 2077	2077	14	40	26
Revenue Act of 2078	2078	14	40	26
Revenue Act of 2079	2079	14	40	26
Revenue Act of 2080	2080	14	40	26
Revenue Act of 2081	2081	14	40	26
Revenue Act of 2082	2082	14	40	26
Revenue Act of 2083	2083	14	40	26
Revenue Act of 2084	2084	14	40	26
Revenue Act of 2085	2085	14	40	26
Revenue Act of 2086	2086	14	40	26
Revenue Act of 2087	2087	14	40	26
Revenue Act of 2088	2088	14	40	26
Revenue Act of 2089	2089	14	40	26
Revenue Act of 2090	2090	14	40	26
Revenue Act of 2091	2091	14	40	26
Revenue Act of 2092	2092	14	40	26
Revenue Act of 2093	2093	14	40	26
Revenue Act of 2094	2094	14	40	26
Revenue Act of 2095	2095	14	40	26
Revenue Act of 2096	2096	14	40	26
Revenue Act of 2097	2097	14	40	26
Revenue Act of 2098	2098	14	40	26
Revenue Act of 2099	2099	14	40	26
Revenue Act of 2100	2100	14	40	26
Revenue Act of 2101	2101	14	40	26
Revenue Act of 2102	2102	14	40	26
Revenue Act of 2103	2103	14	40	26
Revenue Act of 2104	2104	14	40	26
Revenue Act of 2105	2105	14	40	26
Revenue Act of 2106	2106	14	40	26
Revenue Act of 2107	2107	14	40	26
Revenue Act of 2108	2108	14	40	26
Revenue Act of 2109	2109	14	40	26
Revenue Act of 2110	2110	14	40	26
Revenue Act of 2111	2111	14	40	26
Revenue Act of 2112	2112	14	40	26
Revenue Act of 2113	2113	14	40	26
Revenue Act of 2114	2114	14	40	26
Revenue Act of 2115	2115	14	40	26
Revenue Act of 2116	2116	14	40	26
Revenue Act of 2117	2117	14	40	26
Revenue Act of 2118	2118	14	40	26
Revenue Act of 2119	2119	14	40	26
Revenue Act of 2120	2120	14	40	26
Revenue Act of 2121	2121	14	40	26
Revenue Act of 2122	2122	14	40	26
Revenue Act of 2123	2123	14	40	26
Revenue Act of 2124	2124	14	40	26
Revenue Act of 2125	2125	14	40	26
Revenue Act of 2126	2126	14	40	26
Revenue Act of 2127	2127	14	40	26
Revenue Act of 2128	2128	14	40	26
Revenue Act of 2129	2129	14	40	26
Revenue Act of 2130	2130	14	40	26
Revenue Act of 2131	2131	14	40	26
Revenue Act of 2132	2132	14	40	26
Revenue Act of 2133	2133	14	40	26
Revenue Act of 2134	2134	14	40	26
Revenue Act of 2135	2135	14	40	26
Revenue Act of 2136	2136	14	40	26
Revenue Act of 2137	2137	14	40	26
Revenue Act of 2138	2138	14	40	26
Revenue Act of 2139	2139	14	40	26
Revenue Act of 2140	2140	14	40	26
Revenue Act of 2141	2141	14	40	26
Revenue Act of 2142	2142	14	40	26
Revenue Act of 2143	2143	14	40	26
Revenue Act of 2144	2144	14	40	26
Revenue Act of 2145	2145	14	40	26
Revenue Act of 2146	2146	14	40	26
Revenue Act of 2147	2147	14	40	26
Revenue Act of 2148	2148	14	40	26
Revenue Act of 2149	2149	14	40	26
Revenue Act of 2150	2150	14	40	26
Revenue Act of 2151	2151	14	40	26
Revenue Act of 2152	2152	14	40	26
Revenue Act of 2153	2153	14	40	26
Revenue Act of 2154	2154	14	40	26
Revenue Act of 2155	2155	14	40	26
Revenue Act of 2156	2156	14	40	26
Revenue Act of 2157	2157	14	40	26
Revenue Act of 2158	2158	14	40	26
Revenue Act of 2159	2159	14	40	26
Revenue Act of 2160	2160	14	40	26
Revenue Act of 2161	2161	14	40	26
Revenue Act of 2162	2162	14	40	26
Revenue Act of 2163	2163	14	40	26
Revenue Act of 2164	2164	14	40	26
Revenue Act of 2165	2165	14	40	26
Revenue Act of 2166	2166	14	40	26
Revenue Act of 2167	2167	14	40	26
Revenue Act of 2168	2168	14	40	26
Revenue Act of 2169	2169	14	40	26
Revenue Act of 2170	2170	14	40	26
Revenue Act of 2171	2171	14	40	26
Revenue Act of 2172	2172	14	40	26
Revenue Act of 2173	2173	14	40	26
Revenue Act of 2174	2174	14	40	26
Revenue Act of 2175	2175	14	40	26
Revenue Act of 2176	2176	14	40	26
Revenue Act of 2177	2177	14	40	26
Revenue Act of 2178	2178	14	40	26
Revenue Act of 2179	2179	14	40	26
Revenue Act of 2180	2180	14	40	26
Revenue Act of 2181	2181	14	40	26
Revenue Act of 2182	2182	14	40	26
Revenue Act of 2183	2183	14	40	26
Revenue Act of 2184	2184	14	40	26
Revenue Act of 2185	2185	14	40	26
Revenue Act of 2186	2186	14	40	26
Revenue Act of 2187	2187	14	40	26
Revenue Act of 2188	2188	14	40	26
Revenue Act of 2189	2189	14	40	26
Revenue Act of 2190	2190	14	40	26
Revenue Act of 2191	2191	14	40	26
Revenue Act of 2192	2192	14	40	26
Revenue Act of 2193	2193	14	40	26
Revenue Act of 2194	2194	14	40	26
Revenue Act of 2195	2195	14	40	26
Revenue Act of 2196	2196	14	40	26
Revenue Act of 2197	2197	14	40	26
Revenue Act of 2198	2198	14	40	26
Revenue Act of 2199	2199	14	40	26
Revenue Act of 2200	2200	14	40	26
Revenue Act of 2201	2201	14	40	26
Revenue Act of 2202	2202	14	40	26
Revenue Act of 2203	2203	14	40	26
Revenue Act of 2204	2204	14	40	26
Revenue Act of 2205	2205	14	40	26
Revenue Act of 2206	2206	14	40	26
Revenue Act of 2207	2207	14	40	26
Revenue Act of 2208	2208	14	40	26
Revenue Act of 2209	2209	14	40	26
Revenue Act of 2210	2210	14	40	26
Revenue Act of 2211	2211	14	40	26
Revenue Act of 2212	2212	14	40	26
Revenue Act of 2213	2213	14	40	26
Revenue Act of 2214	2214	14	40	26
Revenue Act of 2215	2215	14	40	26
Revenue Act of 2216	2216	14	40	26
Revenue Act of 2217	2217	14	40	26
Revenue Act of 2218	2218	14	40	26
Revenue Act of 2219	2219	14	40	26
Revenue Act of 2220	2220	14	40	26
Revenue Act of 2221	2221	14	40	26
Revenue Act of 2222	2222	14	40	26
Revenue Act of 2223	2223	14	40	26
Revenue Act of 2224	2224	14	40	26
Revenue Act of 2225	2225	14	40	26
Revenue Act of 2226	2226	14	40	26
Revenue Act of 2227	2227	14	40	26
Revenue Act of 2228	2228	14	40	26
Revenue Act of 2229	2229	14	40	26
Revenue Act of 2230	2230	14	40	26
Revenue Act of 2231	2231	14	40	26
Revenue Act of 2232	2232	14	40	26
Revenue Act of 2233	2233	14	40	26
Revenue Act of 2234	2234	14	40	26
Revenue Act of 2235	2235	14	40	26
Revenue Act of 2236	2236	14	40	26
Revenue Act of 2237	2237	14	40	26
Revenue Act of 2238	2238	14	40	26
Revenue Act of 2239	2239	14	40	26
Revenue Act of 2240	2240	14	40	26
Revenue Act of 2241	2241	14	40	26
Revenue Act of 2242	2242	14		

Finding: `str_squish()` recovers substantial matches for many acts (some double their count), but does not rescue the 5 acts with zero matches. The zero-match acts have structural naming problems beyond whitespace.

The Five Unmatched Acts

Table 5: Acts with Zero Tier 2 Matches Even After Whitespace Fix

Act Name	Year
Changes in Depreciation Guidelines and Revenue Act of 1962	1962
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954
Public Law 89-800 (Suspension of Investment Tax Credit)	1966
Public Law 90-26 (Restoration of the Investment Tax Credit)	1967
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1997

Each has a distinct naming problem:

1. **Changes in Depreciation Guidelines and Revenue Act of 1962:** Compound name joining an executive action with a legislative act. Documents reference each component separately.
2. **Expiration of Excess Profits Tax and of Temporary Income Tax Increases:** Descriptive event label, not a formal legislative title. Documents describe the expiration itself.
3. **Public Law 89-800 / 90-26:** Numbered public laws. Documents reference these by their common policy names (“suspension/restoration of the investment tax credit”).
4. **Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997:** Compound name joining two separate acts that passed together.

Subcomponent Decomposition

To handle these compound names systematically, we define a `generate_subcomponents()` function that decomposes any act name into searchable terms. The function applies five strategies: split on “and”, extract parenthetical content, find embedded formal act names (e.g., “Revenue Act of 1962” inside a compound), extract Public Law numbers, and construct action + subject phrases for

event-type names. It also excludes overly broad terms: both topic-level phrases (“excess profits tax”) and generic split fragments (“Revenue”, “Jobs”) that match too widely to identify specific legislation (see Table 7 below).

Table 6 applies the function to the 5 unmatched acts and counts chunk matches for each generated search term.

Table 6: Subcomponent Matches for the 5 Unmatched Acts

Act	Year	Mechanism	Search Term	Matches
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Full name	Changes in Depreciation Guidelines and Revenue Act of 1962	0
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Split 'and' on	Changes in Depreciation Guidelines	0
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Split 'and' on	Revenue Act of 1962	86
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Action subject +	changes of the depreciation guidelines	0
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Action subject +	changes of depreciation guidelines	0
Expiration of Excess Profits Tax	1954	Full name	Expiration of Excess Profits Tax	0
Expiration of Excess Profits Tax	1954	Full name	Expiration of Excess Profits Tax	0
Expiration of Excess Profits Tax and of Temporary Investment Income Tax (Credit) Increases	1962	Full name and of Temporary Investment Income Tax (Credit) Increases	Expiration of Excess Profits Tax and of Temporary Investment Income Tax (Credit) Increases	130

Key findings:

- **4 of 5 acts become findable** through subcomponent decomposition. Split on “and” recovers formal legislative names (“Revenue Act of 1962” with 56 matches; “Taxpayer Relief Act of 1997” with 20; “Balanced Budget Act of 1997” with 70). Parenthetical extraction finds “Suspension of the Investment Tax Credit” (9 matches) and “Restoration of the Investment Tax Credit” (6 matches) for the Public Law acts.
- **Public Law numbers** (89-800, 90-26) provide additional specific matches (2-3 chunks each).
- **The Expiration of Excess Profits Tax act remains at 0 matches.** The action + subject mechanism produces malformed phrases (“expiration of the of excess profits tax”) due to the unusual “of...and of” structure. This act requires a different approach, explored in the next section.

Proposed Matching Improvements

Based on these findings, we can improve positive identification with changes that stay at chunk level throughout. This section addresses the first two; a third mechanism (co-occurrence matching) emerges from the excess profits investigation in the next section.

1. Whitespace normalization before matching

Apply `str_squish()` to both act names and chunk text before `str_detect()`. This collapses OCR line breaks without changing the matching logic. No new dependencies, no mixed granularities.

Expected impact: Recovers additional matches for most acts (29 of 40), with some doubling their match count.

2. Subcomponent matching for compound act names

Apply `generate_subcomponents()` (defined above) to decompose compound act names into searchable terms. As demonstrated in Table 6, this recovers the 5 previously unmatched acts.

3. Search terms to exclude from Tier 2

The decomposition and audit analysis identified two categories of overly broad terms that `generate_subcomponents()` excludes. **Topic-level phrases** match policy areas rather than specific acts. **Split fragments** are single words or short phrases produced by splitting compound act names on “and” that are too generic to identify any legislation.

Table 7: Search Terms Excluded from Tier 2 as Overly Broad

Excluded Phrase	Category	Problem	Matches
revenue	Split fragment	Common fiscal term appearing in nearly every document	8042
economic growth	Split fragment	Generic macroeconomic phrase	3278
jobs	Split fragment	Generic labor-market term	2956
balanced budget	Topic phrase	Matches general budget balance commentary	967
tax reduction	Split fragment	Matches any tax cut discussion	945
excess profits tax	Topic phrase	Matches Korean War era and WWII discussions broadly	630
job creation	Split fragment	Generic labor-market phrase	544
investment tax credit	Topic phrase	Matches every ITC discussion across 80 years	402
tax equity	Split fragment	Matches any tax fairness discussion	119
taxpayer re- depreciation in- terim	Topic phrase	Matches any phrase used in many policy contexts	106
guidelines	Topic phrase	Matches any phrase used in many policy contexts	72

For compound-name acts, we rely on the remaining specific subcomponents: formal legislative names (“Revenue Act of 1962”, “Balanced Budget Act of 1997”), parenthetical descriptions (“Suspension of the Investment Tax Credit”), and Public Law numbers.

Table 8: Cumulative Impact of Proposed Matching Improvements

Strategy	Total Matches	Unmatched Acts
Current (exact match)	1,219	5
1. Whitespace normalization	1,738	5
1. Subcomponent matching	2,005	1

Each strategy builds on the previous.

Table 9: Per-Act Chunk Matches Under Proposed Strategy

Act	Year	Matching Chunks
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1997	138
Economic Growth and Tax Re- lief Reconciliation Act of 2001	2001	122
Tax Reform Act of 1986	1986	111
Economic Recovery Tax Act of 1981	1981	107
Jobs and Growth Tax Relief Reconciliation Act of 2003	2003	106
Tax Reform Act of 1976	1976	89
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	86
Revenue Act of 1978	1978	84
Tax Reform Act of 1969	1969	77
Revenue Act of 1950	1950	71
Omnibus Budget Reconcilia- tion Act of 1993	1993	70
Revenue Act of 1964	1964	68
Omnibus Budget Reconcilia- tion Act of 1990	1990	67
Social Security Amendments of 1983	1983	66
Revenue Act of 1971	1971	64
Tax Reduction Act of 1975	1975	60
Tax Equity and Fiscal Respon- sibility Act of 1982	1982	58
Revenue Act of 1945	1945	49
Deficit Reduction Act of 1984	1984	46
Revenue Act of 1948	1948	43
Job Creation and Worker Assis- tance Act of 2002	2002	41
Expiration of Excess Profits Tax Act of 1969	1969	38
Revenue Act of 1958	1958	38
Revenue Act of 1950 (Tax credit) and Revenue Act of 1958 Tax Increases	1950	31

Search term audit

The tables below list every search term that `generate_subcomponents()` produces across all 40 acts. Table 10 shows each term with its chunk match count (orange rows exceed 100 matches and warrant manual review). Table 11 pivots the same data by mechanism, so we can see which decomposition strategy drives matches for each act.

Table 11: Per-Act Chunk Matches by Matching Mechanism

[illegible]

Finding the Excess Profits Tax Expiration

The act “Expiration of Excess Profits Tax and of Temporary Income Tax Increases” is the hardest to match via subcomponent decomposition. The full name yields 0 matches, the split-on-“and” terms are fragile, and the action+subject mechanism produces malformed phrases. Meanwhile “excess profits tax” alone was excluded as overly broad (336 matches). But how broad is it really? If those matches cluster around the Korean War era, “excess profits tax” may be specific enough in practice.

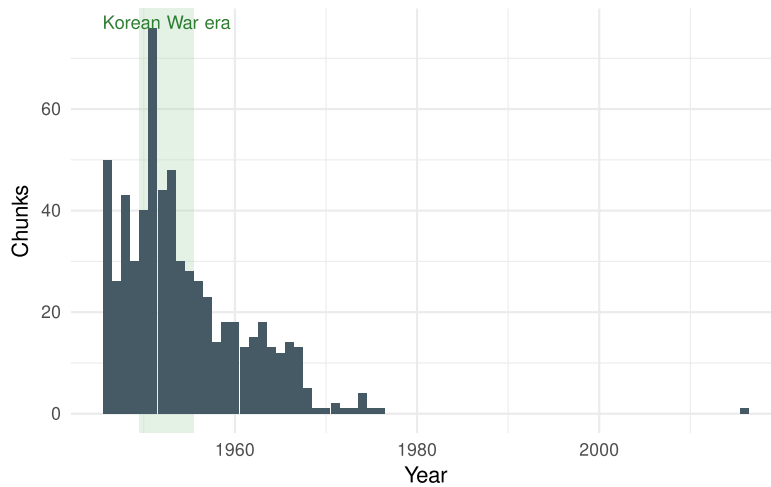


Figure 1: Chunks matching “excess profits tax” by document year. The Korean War era (1950–1955, shaded) accounts for the majority of mentions.

Table 12: Decade Summary: “excess profits tax” Mentions

Decade	Chunks
1940s	149
1950s	347
1960s	122
1970s	11
2010s	1

Figure 1 and Table 12 show that mentions cluster heavily in the Korean War era, but the term persists across decades. Regardless of concentration, the term alone is too generic: it matches any discussion of the policy area, not the specific 1953 expiration event. The strategy comparison below confirms that **co-occurrence** (“expiration” AND “excess profits” in the same chunk) is the right

approach: it captures the policy event precisely without requiring exact substring ordering, making it robust to OCR artifacts and varied phrasing.

Table 13: Matching Strategies for “Expiration of Excess Profits Tax”

Strategy	Chunks
"excess profits tax" (all years)	630
"expiration" AND "excess profits" (co-occurrence)	52
"expiration of excess profits tax" (exact substring)	0

Decision: For “Expiration of Excess Profits Tax and of Temporary Income Tax Increases”, the pipeline will use **co-occurrence matching**: a chunk is positive if it contains both “expiration” and “excess profits”. This introduces a new matching mechanism alongside substring matching and subcomponent decomposition. The `identify_tier2_chunks()` function will need a co-occurrence lookup table for acts where single-term matching is insufficient.

Timing Consistency Across All Acts

The excess profits investigation above examined one act’s temporal distribution. This section generalizes the analysis to all 44 acts, asking: **do matching chunks come from documents published near the act’s year, or are matches scattered across the full corpus?** Temporal clustering around the act year is a sign of precise matching; a flat distribution signals that the search terms are too generic.

We compute two views. First, the raw document-year distribution for each act’s matching chunks. Second, for acts with a year in their name, the difference between the chunk’s parent document year and the year embedded in the act name. A concentration around zero in the difference plot means the matching chunks come from contemporaneous documents.

Raw chunk-year distributions

Figure 2 shows the document-year distribution of matching chunks for each act. The red dashed line marks the act’s signed year from `aligned_data`. Acts with tight clustering around the red line have precise, temporally specific matches. Acts with dispersed distrib-

utions (e.g., those containing common policy terms) may need tighter matching criteria.

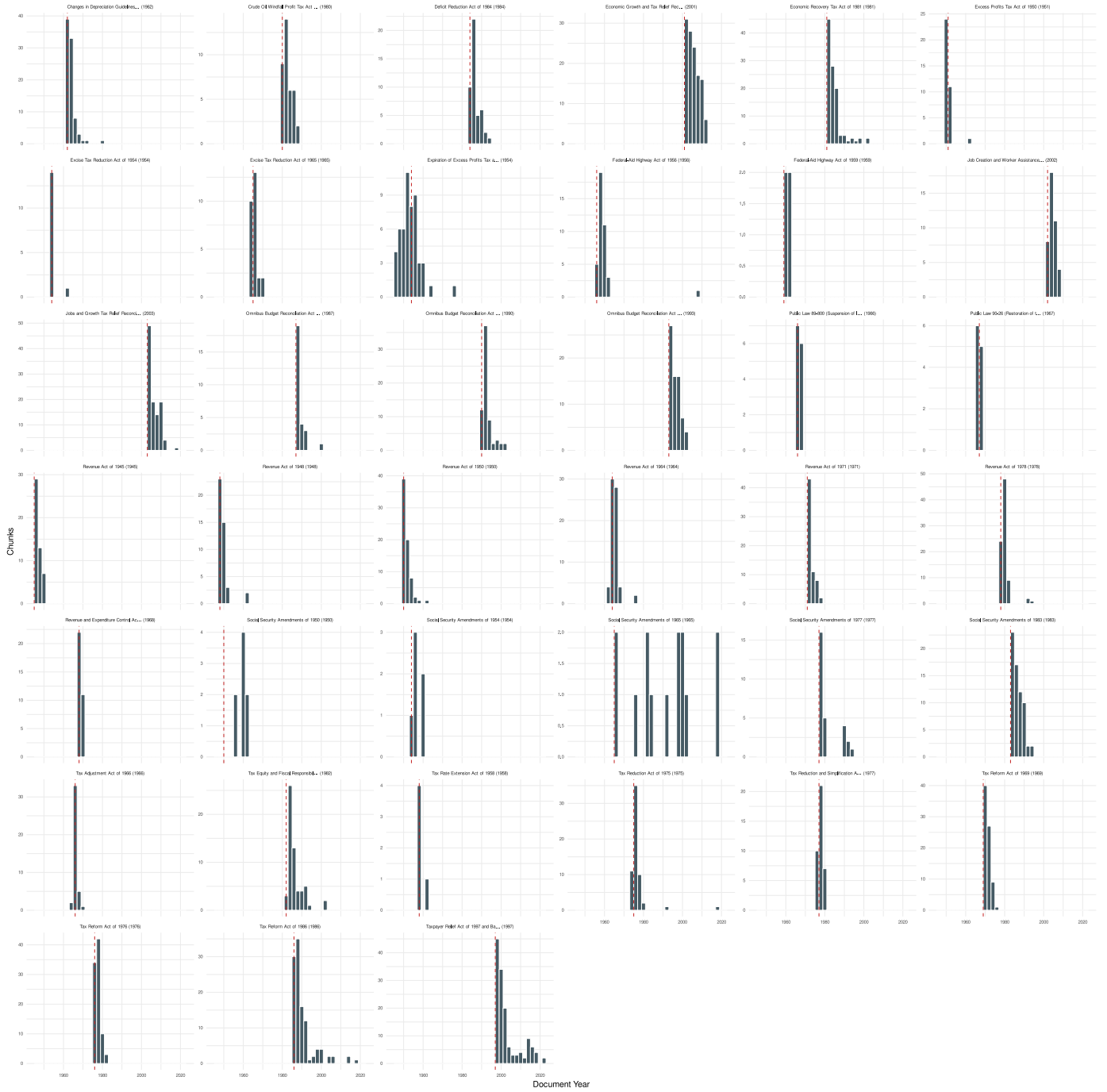


Figure 2: Document-year distribution of matching chunks for each known act. Red dashed line marks the act's signed year.

Year-difference distributions

For the subset of acts with a year in their name (e.g., “Revenue Act of **1962**”), Figure 3 plots the difference between each matching chunk’s document year and the year embedded in the act name. A spike at zero means most matches come from the same year the act was enacted. Positive values indicate retrospective discussion in later documents (common in Economic Reports of the President). Negative values would indicate pre-enactment discussion.

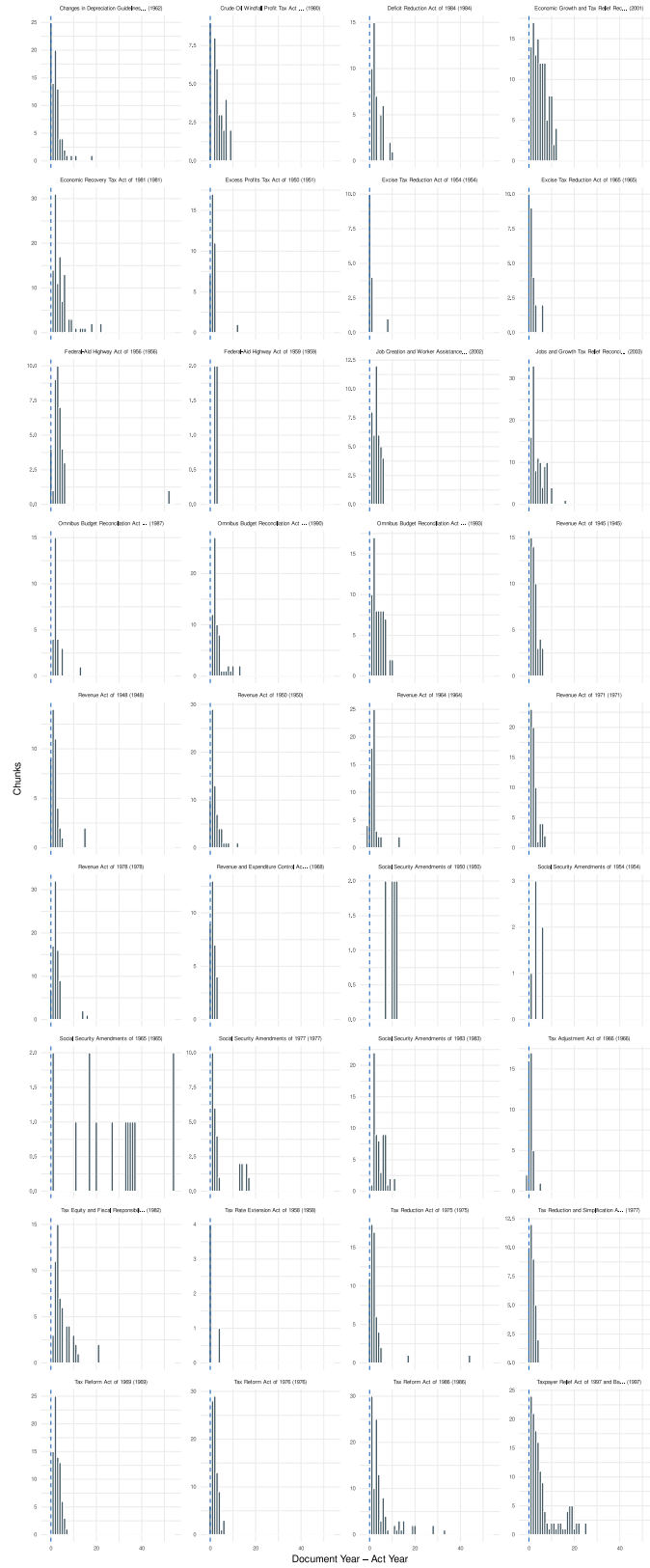


Figure 3: Distribution of (document year – act name year) for matching chunks. Only acts with a year in their name are shown. Blue dashed line marks zero (contemporaneous).

Act	Signed	Name	Year	Chunks	Median Diff	Within ±2 yr	Within ±5 yr
Social Security Amendments of 1950	1950	1950	8	10.5	0.0%	0.0%	
Social Security Amendments of 1965	1965	1965	14	30.0	14.3%	14.3%	
Social Security Amendments of 1954	1954	1954	6	3.0	16.7%	66.7%	
Tax Equity and Fiscal Revenues Act of 1982	1982	1982	58	3.5	24.1%	72.4%	
Changes to Economic Recovery Tax Act of 1981	1981	1981	122	5.0	25.4%	100.0%	

by within ± 2 year percentage (ascending). Green
orange 25–50%, red <25%.
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Acts with low temporal concentration (red rows in Table 14) are candidates for tighter matching: their search terms may be matching generic policy discussion rather than the specific legislation. Acts with high concentration (green rows) confirm that our proposed matching strategy identifies the right documents.

Rethinking Negatives

With improved positive matching, we can also relax the negative criterion. The current system excludes all chunks matching `relevance_keys` from the negative pool, leaving only 35 clean negatives (1.2% of all chunks). This makes precision untestable.

Proposed: Include all non-positive chunks as negatives, and compute a continuous **key density** score ($n_relevance_keys / n_words$) that measures how much fiscal vocabulary each negative chunk contains. This replaces the binary easy/hard distinction with a gradient:

- Low density (e.g., < 1%): general economic forecasts, administrative content
- Medium density (e.g., 1-5%): economic commentary with some fiscal vocabulary
- High density (e.g., > 5%): substantive fiscal policy discussion without naming a specific act

This continuous measure feeds directly into S3 error analysis: we can plot the model's false positive rate against key density to identify the threshold where discrimination breaks down. No arbitrary cutoff needed at the data preparation stage.

Table 15: Proposed Tier System with Expanded Negative Pool

Category	N	Evaluation Role
Tier 1 (verbatim passage)	70	Gold standard positive
Tier 2 (act name match)	1410	Inferred positive
Negatives (no act match, no relevance keys)	11632	Negative (low key density expected)
Negatives (no act match, has relevance keys)	1949	Negative (higher key density expected)
Total	15061	

Under this scheme, `prepare_c1_chunk_data()` will compute `key_density` for every negative chunk. During S3 error analysis, we can stratify false positives by density bins to understand whether the model struggles with fiscal-vocabulary-rich negatives (expected) or also fails on clean negatives (a more serious problem).

Next Steps

1. **Implement whitespace normalization** in `identify_tier2_chunks()`: apply `str_squish()` to both act names and chunk text before matching
2. **Implement subcomponent matching** for compound act names: split on " and ", extract parenthetical descriptions, find embedded formal act names. Exclude the 5 overly broad phrases identified above.
3. **Implement co-occurrence matching** for event-type acts: a lookup table mapping acts to term pairs (e.g., "expiration" + "excess profits") for cases where substring matching is insufficient
4. **Relax negative criterion**: remove `relevance_keys` exclusion from negatives. Compute `key_density` ($n_relevance_keys / n_words$) as a continuous measure for each negative chunk. This feeds into S3 error analysis.
5. **Validate coverage**: confirm all 44 acts have at least one Tier 2 match after improvements
6. **Update `prepare_c1_chunk_data()`** to return the revised tier assignments and key-density-tagged negatives

References

Romer, Christina D., and David H. Romer. 2010. “The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks.” *American Economic Review* 100 (3): 763–801.